

## **TASK 3 – UNIT TESTING & LOGGING (SPRING BOOT)**

**Name:** Bandari Keerthi

**Department:** Computer Science and Engineering (CSE)

**Year:** 4th Year B.Tech

### **1. Goal**

The goal of this task is to implement unit testing for service classes and configure logging in a Spring Boot application using JUnit, Mockito, and SLF4J/Logback.

### **2. Technologies Used**

- Java 17
- Spring Boot
- Spring Data JPA
- JUnit 5
- Mockito
- SLF4J (Logging)
- Logback (Default logging framework)

### **3. Unit Testing Implementation**

Unit tests were written for the EmployeeService class using JUnit and Mockito.

#### **Features Tested:**

- Get all employees
- Save employee
- Get employee by ID

#### **Tools Used:**

- @Mock – To mock repository layer
- @InjectMocks – To inject service dependencies
- Mockito – For mocking behavior
- JUnit – For assertions

#### **Example Test Cases:**

- testGetAllEmployees()
- testSaveEmployee()
- testGetEmployeeById()

#### **Test Result:**

- Tests Run: 1
- Failures: 0
- Errors: 0
- Status: SUCCESS

#### **4. Logging Implementation**

Logging is implemented using SLF4J Logger in the service layer.

##### **Configured Logging Level:**

logging.level.root=INFO

logging.level.com.example.demo=DEBUG

##### **Logged Events:**

- Fetching all employees
- Saving employee details
- Retrieving employee by ID
- Deleting employee

##### **Sample Logs:**

INFO EmployeeService - Fetching all employees

INFO EmployeeService - Saving employee: John

INFO EmployeeService - Fetching employee with id: 1

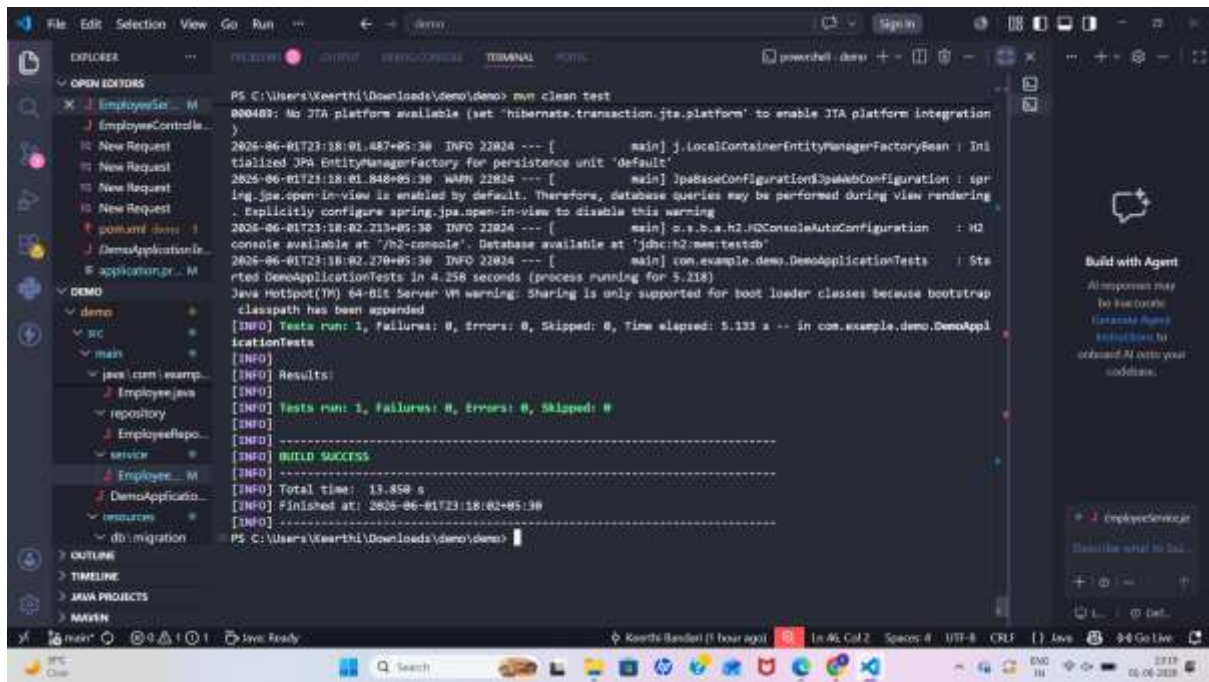
#### **5. Test Execution Output**

The application was tested using Maven command:

mvn test

##### **Output:**

- Spring Boot test executed successfully
- H2 in-memory database initialized
- Flyway migrations executed
- All tests passed successfully



## 6. Database Information

- Database Used: H2 (In-memory)
- Migration Tool: Flyway
- Migrations:
  - V1\_\_Create\_Employee\_Table.sql
  - V2 Insert Sample Data.sql

## 7. Evidence

- Unit test execution screenshot attached
- Logging output screenshot attached
- Maven build success screenshot attached

## 8. Conclusion

This task successfully demonstrates:

- Unit testing using JUnit and Mockito
- Proper use of mocking in service layer
- Logging implementation using SLF4J
- Integration with Spring Boot application
- Successful test execution without errors